NORTH BENGAL INTERNATIONAL UNIVERSITY

Lab Report : 7

TOPIC : WRITE AN ASSEMBLY CODE TO SHOW BRANCHING
COURSE CODE : CSE 0611-3322
COURSE TITLE : MICROPROCESSOR & ASSEMBLY LANGUAGE SESSIONAL.

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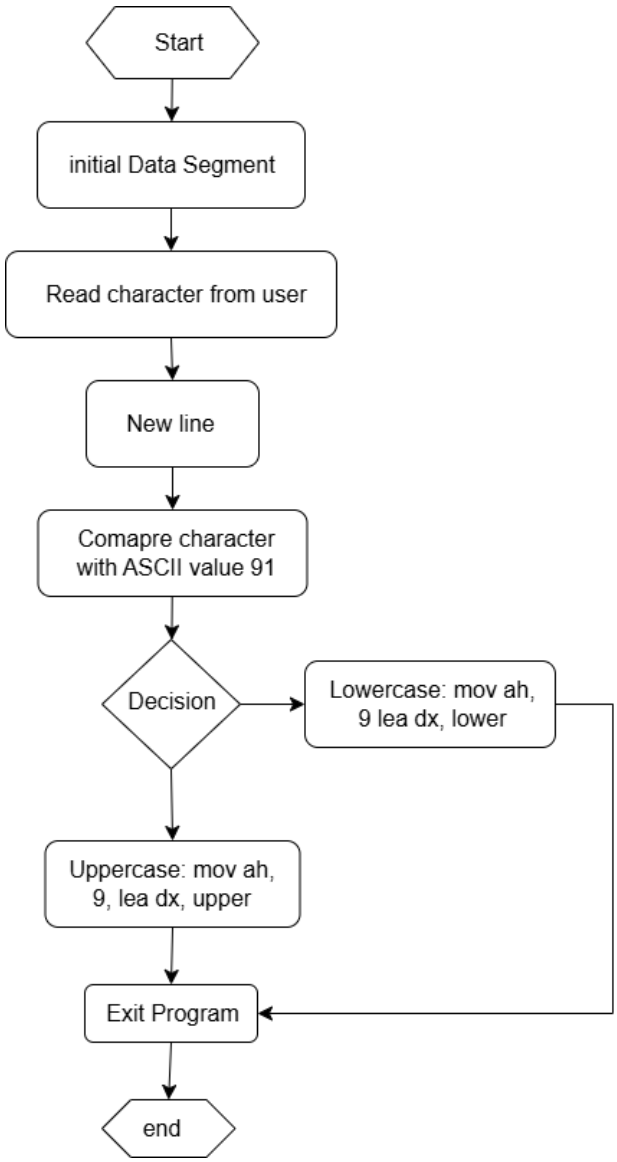
Theory:

Flowchart of Branching:



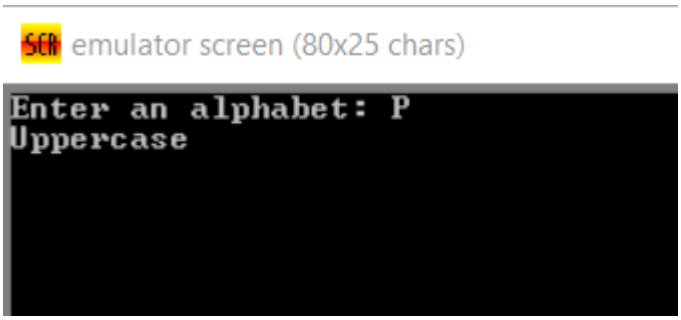
What is Branching:

Branching in assembly language involves altering the flow of execution based on certain conditions. It uses instructions like `jmp` (unconditional jump) and conditional jumps (`je`, `jne`, `jg`, `jl`, etc.) to direct the program to different parts of the code. This allows for implementing control structures such as if-else statements and loops, enabling more complex and flexible program logic by responding to specific conditions during execution.

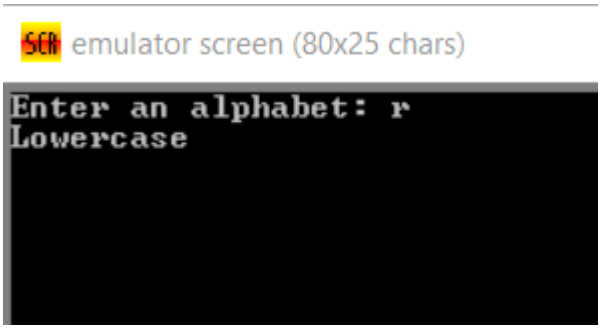
Code:

```
.model small
.stack 100h

.data
    input db 'Enter an alphabet: ', '$'  ; Prompt message
    upper db 'Uppercase', 0ah, 0dh, '$'  ; Uppercase message
    lower db 'Lowercase', 0ah, 0dh, '$'  ; Lowercase message

.code
main proc
    ; Initialize data segment
    mov ax, @data
    mov ds, ax

    ; Prompt user for input
    mov ah, 9
    lea dx, input
    int 21h

    ; Take input
    mov ah, 1
    int 21h
    mov bl, al

    ; New line
    mov ah, 2
    mov dl, 0ah
    int 21h
    mov dl, 0dh
    int 21h

    ; If-else to check case
    cmp bl, 91
    jge level_low
    jmp level_up

level_up:
    ; Display uppercase message
    mov ah, 9
    lea dx, upper
    int 21h
    jmp exit

level_low:
    ; Display lowercase message
    mov ah, 9
    lea dx, lower
    int 21h

exit:
    ; Exit program
    mov ah, 4ch
    int 21h

main endp
end main
```

Output-1:



Output-2:



Conclusion: This code demonstrates how to prompt a user for input and determine whether the input is an uppercase or lowercase letter in assembly language. By utilizing DOS interrupts, it effectively handles user input and displays appropriate messages based on the character's case. Through this practice, we've gained insights into setting up data segments, using conditional jumps, and handling character I/O operations in assembly language.